

03: Collecting User Data II:

- Query techniques
- Recruitment + Ethics

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User-Centred System Design

Dr Tracey Booth (she/her)

Centre for HCI Design | Computer Science | SST
tracey.booth.2@citystgeorges.ac.uk

Session 3: Learning objectives



- By completing this session, you should be able to:
 - Design and conduct effective data collection using **interviews** and **questionnaires**.
 - Identify and recruit appropriate **participants** for user research activities.
 - Understand the need for treating participants ethically, and know how to obtain **informed consent** from participants.

Tutorial: Direct Naturalistic Observation

Exercise recap



The brief, in brief:

- The team behind the City Nav app have enlisted you to generate some **user insights to inform the next iteration of the app**.
- To help them come up with ideas for the next version of the app, **they want to understand how staff and students use various spaces around the University**.
- As part of your user research, you have decided to conduct **direct naturalistic observations** within these spaces.

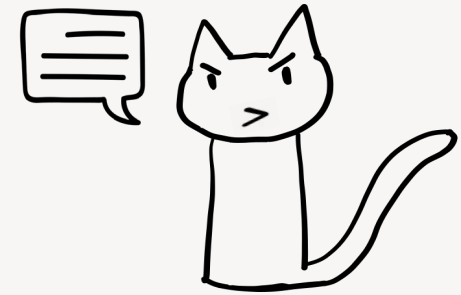
Instructions:

Working in small groups:

- Prepare a template for taking notes.
- Conduct an observation in one or two of the spaces listed.
- Report back via the Session 2 Moodle forum

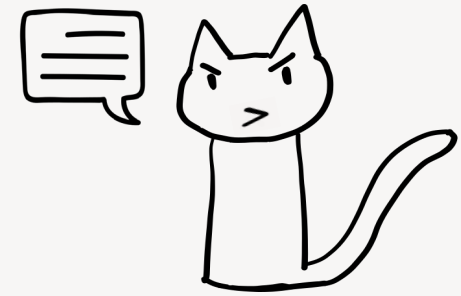
Tutorial: Direct Naturalistic Observation

Debrief



Report findings from your observation(s) to the Session 2 tutorial forum on Moodle, and reflect on the method.

- **Observer(s).** Who was in your group?
- **Observation decisions.** E.g. What area(s) did you observe? What template did you create/use and why? What did you decide to look for and record? What did you decide to ignore? Did you divide the work?
- **Results.** Not a full report, just a few highlights, relevant to the brief.
- **Reflection.** How did you find this exercise and what will you take away from it, e.g. in terms of learning? Thoughts on the technique?



Tutorial: Direct Naturalistic Observation

Prompts for reflection/reporting

- What were your take-aways from doing the tutorial exercise?
- Thoughts in general about the technique?
- Thoughts about specific decisions that you made (areas, template, focus, division of work etc.)
- Anything you found challenging?
- Anything you would do differently or be mindful of in future?
- Would you use it? Under what circumstances? How could you maximise its effectiveness?

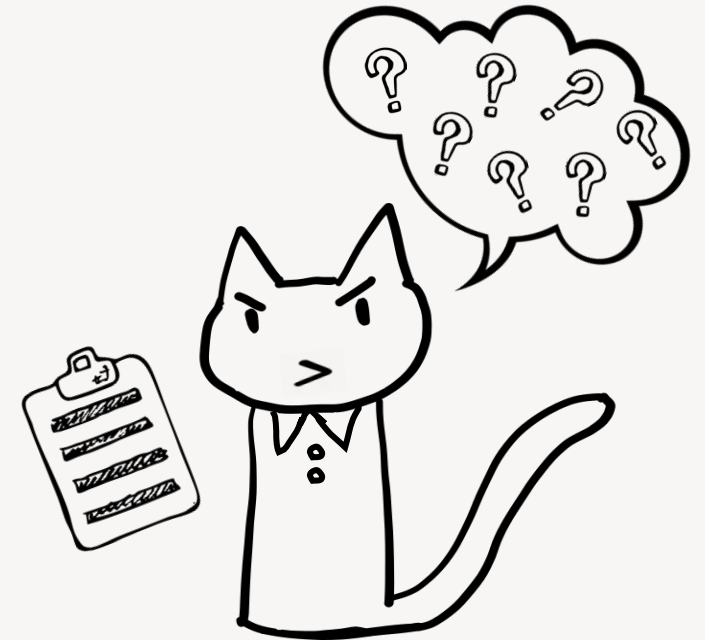
Questions?



Collecting User Data:

Query techniques

- Interviews
- Questionnaires



Query techniques

Self-reporting techniques:

- The researcher asks questions and records the participants' responses.

Questions are designed to elicit:

- **Facts** — what people know.
- **Behaviour** — what people do.
- **Beliefs / attitudes / opinions** — what they think or feel.

e.g. Interviews, Questionnaires/Surveys, Focus Groups.

What do you want to know?

Facts

- *“What do you do for a living?”*
- *“What languages do you speak?”*
- *“How tall are you?”*
- *“What is your highest level of educational qualification?”*
- *“Where do you live?”*
- *“What type of phone do you own?”*
- *“How old are you?”*
- *“What subjects do you study at university?”*
- *“How much do you earn?”*

What do you want to know?

Behaviour

- *“Where do you shop for groceries?”*
- *“Do you donate money to charity, and if so, how frequently?”*
- *“Tell me about your hobbies.”*
- *“What does your typical weekday look like?”*
- *“How often do you attend lectures?”*
- *“Do you smoke?”*
- *“How much alcohol do you drink per week?”*
- *“Describe in detail how you usually travel to university.”*
- *“When do you choose to shop online, rather than going to a store?”*

What do you want to know?

Opinions / beliefs / attitudes

- *“What do you think of the homepage?”*
- *“What is your favourite type of food?”*
- *“How difficult or easy was the task for you?”*
- *“Which of these designs do you prefer, if any, and why?”*
- *“To what extent do you agree/disagree with the statement: ‘I find Moodle very easy to use’ ...?”*
- *“Would you recommend this app to a friend?”*
- *“What do you think of the current Prime Minister?”*
- *“Describe your ideal learning environment.”*

Watchpoint: **under/over-reporting**

- Be mindful that answers may not always reflect the truth.
- Questions can sometimes be seen as threatening:
 - *“I want to respond truthfully but if I tell the truth, you will think I’m a bad person”*
- Topics that people may **over-report**:
 - Being a good citizen, seat belt usage, being well-informed, charitable giving, voting...
- Topics that people may **under-report**:
 - Illness or disabilities, illegal behavior, financial status, use of alcohol...

Interview and questionnaire **design**

Be very clear about your **purpose**: what is your **goal**?

- What do you want to know? Facts? Behaviour? Opinions etc?
- Why do you want to know it? What do you want/need that information?
- What can you ask your respondents to answer those questions?

Don't write specific questions until you have thought through what you will do with the answers.

- For any question you write ask *“Why do I want to know this?”*
 - ! *“It might be interesting to know”* is not an acceptable answer.

“Why do I want to know these things?”

- Name
- Age
- Gender
- ...
- What else?

Ask **specific** questions

Not specific: *“How do you use the City website?”*

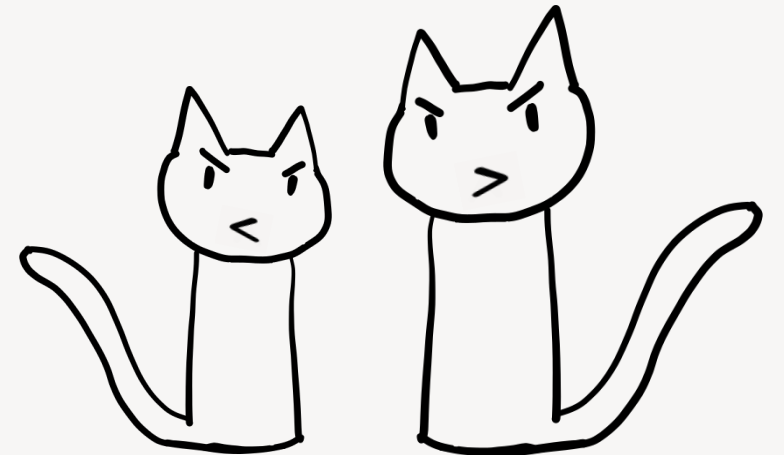
Specific: *“What did you do on the City website, the last time you used it?”*

Questions: **wording and structure**

- Use **simple** vocabulary.
- Use **unambiguous** phrasing.
- Avoid **double** (conjoined) questions.
 - *“What do you think of the City website, and how do you find the layout?”*
- Avoid **loaded** (and other **leading**) words and phrasing.
 - *“Do you think it was a mistake for City University to rebrand?”*
 - *“How much do you love the new logo?”*
- Start with a **reassuring** question that is easy to answer.
- *But* make sure to ask the important things early enough.

Your turn

- *“How good is Instagram?”*
- *“Which newspaper do you read?”*
- *“How old are you?”*
- *“How much do you trust and enjoy using TikTok?”*
- *“How effective is Moodle?”*
- *“How many hours a day do you spend online?”*



Open-ended vs closed-ended questions

Types of questions:

- Open-ended

No restrictions on response

- Closed-ended (or fixed alternative)

Respondent must select one of several fixed alternatives

- Scale

A variant of closed-ended, where respondent must respond on a pre-determined scale e.g. rating the extent to which they agree or disagree with something.

NN/g

[https://www.youtube.com/
watch?v=LpV3tMy_WZ0](https://www.youtube.com/watch?v=LpV3tMy_WZ0)

Open vs. Closed Questions

Open-ended vs closed-ended questions

Open-ended questions

- Good for eliciting rich detail.
- Can reveal unanticipated info.
- Very useful in early design
- Can also be useful in evaluation
- Require more effort to answer—this can be problematic in a questionnaire.
- Data captured is more difficult to analyse.

Closed-ended questions

- Answers aren't very rich in detail.
- Require certainty about the range of possible answers—inflexible
- Not so useful in early stages of design
- More useful in evaluation, e.g. for rating or measuring things.
- Useful for comparison.
- Use [simple] statistics to analyse.

Questions?





Query techniques

Interviews

Interviews

- A conversation “with purpose”.
- Initiated by the interviewer to obtain research-relevant information.
- Useful for obtaining information difficult to elicit using other approaches.
 - e.g. general principles, background knowledge, reasons for behaviour, infrequent events, attitudes...
- Helpful in understanding how a user thinks/feels.
- Less good for capturing what they do—responses may not be complete.
- Generate rich, qualitative data.
- Time-consuming to conduct, transcribe and analyse, but can be more flexible than a questionnaire.
- Danger of researcher bias towards own knowledge and beliefs, raising problems with reliability.

Interview **types**

Three different types:

- **Structured interview**
 - Pre-determined questions; asked in fixed order; like a verbal questionnaire.
- **Semi-structured interview**
 - Questions determined in advance, but may be reworded, reordered, explained, elaborated or even omitted.
- **Unstructured interview**
 - No pre-determined questions; interviewer has a general area of interest, but conversation may develop freely.

Planning and conducting an interview

- A semi-structured interview should be well-planned
- Create and use an **interview guide**.
- Write a **lead statement**, with goals of interview, major topics, and time required—give participant an idea of what to expect.
- **Warmup questions**: Put participant at ease, get them talking, gain some background information, and establish that you are interested.
- Think about topic/question order for a **logical flow**—you may not always stick to it, but it's useful to consider up front.
- Close with a **generic open question**
 - *“Is there anything else you want to tell me?”*

Tips for **effective** interviewing

- Ask straightforward, succinct, non-jargon, non-threatening questions.
- Avoid leading questions or giving other cues to desired answers.
- Don't pre-empt answers, e.g. *“you seem to like this colour...”*
- Acknowledge—be interested, but remain neutral
 - *“uh-huh”, “OK”, “I see”, not “I agree”, or “that dirt bag”*
- Be mindful of your own non-verbal behaviour, e.g. facial expressions and body language, and any messages it might send.
- Use gentle but neutral probing to get them to expand on their answers.
 - *“tell me more about...”*
- Paraphrasing—check understanding
- Focusing, redirecting—*“I'd like to return to...”*
- Keep things grounded in the specific—*“Show me...”*
- Slow down, and allow for silence.

Questions?





Query techniques

Questionnaires

Questionnaires

- A self-reporting technique where people fill in the answers to questions themselves.
- Generally efficient for the researcher (and therefore tempting!)
- Can be online, using survey software.
 - You must use [Qualtrics](https://cityunilondon.eu.qualtrics.com) for any online questionnaires relating to university projects (<https://cityunilondon.eu.qualtrics.com>)
 - Software often has built-in analysis tools.
- Can be paper-based.
 - You will need to do the data entry manually, for analysis.

Seek **specific** answers

How often did you use TikTok during the last month?

- | | |
|---|---|
| ☒ <i>Never</i> | ☐ <i>Not at all</i> |
| ☒ <i>Rarely</i> | ☐ <i>1-3 times a month</i> |
| ☒ <i>Occasionally</i> | ☐ <i>Once a week</i> |
| ☒ <i>Regularly</i> | ☐ <i>Two to four times a week</i> |
| | ☐ <i>Once a day</i> |
| | ☐ <i>More than once a day</i> |

But watch out for *overlapping* categories—don't confuse people!

Open-ended vs closed-ended questions on a questionnaire

- Open-ended questions may require some effort to reply in writing.
- Open-ended questions rely on **recall**:
 - *“Which websites do you visit frequently?”*
- Closed-ended questions with checkboxes can take advantage of **recognition**—but you need to know the possible answers:
 - *“Which of the following websites do you visit at least once a week?”*
 - Include *“other”*—also asking them to specify.
 - *“Not applicable”* may also be appropriate

Closed-ended questions: **measurement scales**

- Nominal
- Ordinal
- Interval
- Ratio

See: <http://www.mymarketresearchmethods.com/types-of-data-nominal-ordinal-interval-ratio>

Measurement scale **data types**

- **Nominal** : Categories with no order.
 - e.g., Gender, Type of device used
- **Ordinal**: Categories with a meaningful order, but uneven intervals.
 - e.g., Satisfaction ratings: Poor, Fair, Good, Excellent.
- **Interval**: Ordered values with equal intervals, but no true zero.
 - e.g., Temperature in °C
- **Ratio**: Like interval, but with a true zero, allowing meaningful ratios.
 - e.g., Age, income, time spent online.

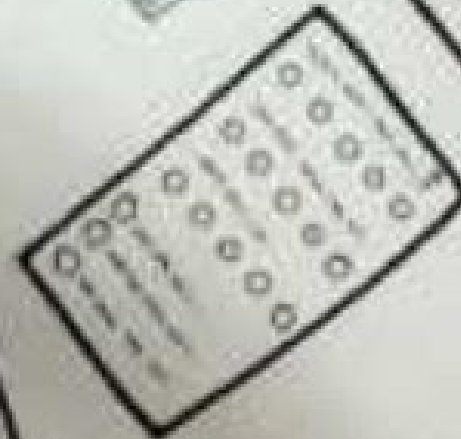
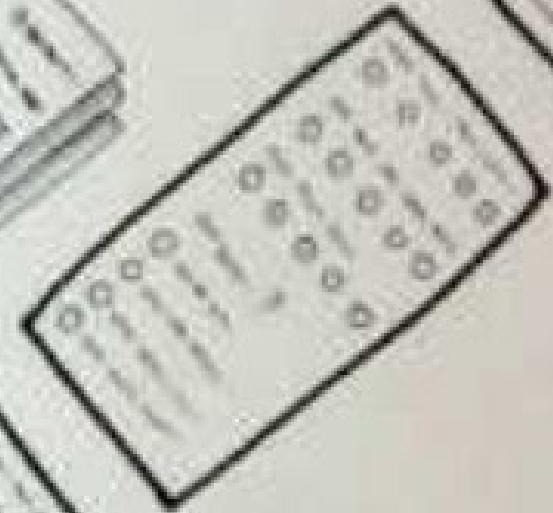
Ways to **analyse** questionnaire data

- Scaled responses:
 - Calculating the means, doing t-tests
- Categories:
 - Calculating frequencies, doing chi-squares.
- Open questions:
 - Code answers and group into categories (this is qualitative data analysis).

GRAY

Avoid double negatives

Avoid questions with two concepts



[http://uxmastery.com/
better-user-research-
through-surveys](http://uxmastery.com/better-user-research-through-surveys)

Questionnaires: **Design? Adopt?**

- You are not the first to do this: others have developed questionnaires on almost everything.
- Some have even **validated** the questions

“A questionnaire is considered validated if the questionnaire has been shown to have a high reliability and internal consistency.

A questionnaire with a high reliability would elicitate the same answers if applied to the same population again.

Internal consistency is measured by comparing the answers to questions measuring the same concepts.”

(Field Epidemiology Manual)

Questions?





Some **general points** about query techniques

Asking the right people

- ‘Sampling’
 - Who is your target population?
 - Sample size.
 - For quantitative data, more is better
 - The greater variation among people, the more you have to ask
 - Getting the people.
 - How will you find them (ethically)?
- ⚠ Not asking the right people can result in misleading data

Interviews & questionnaires: **strengths**

- Faster to carry out than observations.
- Can cover infrequent events.
 - e.g. something that happens just once a year).
- Can cover activities that are difficult to observe.
- Can deliver quantitative data from closed questions (this may or may not be useful).

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Interviews & questionnaires: **drawbacks**

- Information is an idealised version of what should happen rather than what does happen, often generalised and lacking detail.
- Responses may lack accuracy or honesty.
- Danger of researcher bias towards subset of knowledge that *they* possess— you only find out about what you ask about.

Interviews & questionnaires:

Uses in user-centred system design

- Interviews can be useful during initial user research to help you find out about users, what they do, where they do it, what their experience is, what difficulties they encounter...
- Questionnaires have limited value during initial user research. They're often better to use in evaluation.
- In both cases, you cannot possibly predict the possible answers in advance. You need some open-ended questions!

Query techniques: **warning!**

- The user is the user.
- The designer is the designer.
- Do not expect the user to tell you what the system should do.
- Don't just ask the user what they want.

You can't *always* trust what you hear...

- People may not accurately report on:
 - Decision-making.
 - Complex problem-solving (insight).
 - Changes in opinion.
 - Detailed, skilled behaviour.
 - Embarrassing facts.
- Verbal reports are **reconstructions**, not direct reports.

Recipe for using query techniques

1. Decide what information is needed—what is the **goal**?
2. Decide **who** you will ask.
3. Decide what **method** of asking.
4. Conduct a search for **existing** questions and/or scales.
5. Draft **new** questions or revise existing ones.
6. Put questions in **sequence**.
7. If a questionnaire, **format** it.
8. Get peers to **evaluate**.
9. **Pilot** on a few (if it's a questionnaire, get them to think out loud).
10. Fix any issues, then use it!

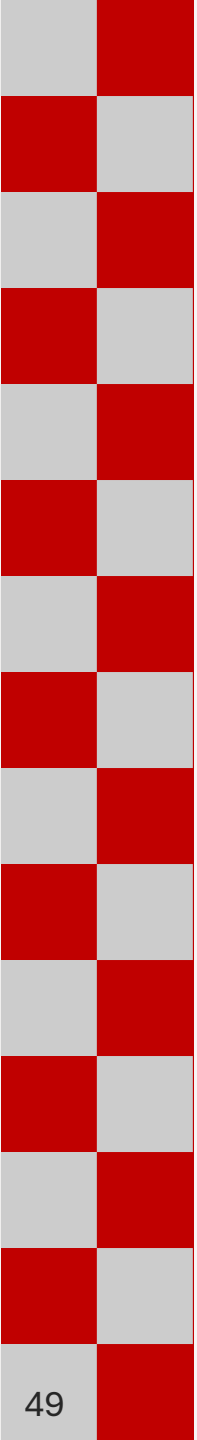
Tips for interviews and questionnaires

(Useful for checking the design of your interview/questionnaire)

- Ask **specific** questions.
- Use **straightforward, unambiguous** language.
- Avoid **jargon**.
- Don't ask **double** questions.
- Avoid **biased/leading** questions and **loaded** words.
- Start with an **easy, reassuring** question.
- Get to the **important** things quickly.
- Use open-ended questions to gather **rich detail**.
- Watch out for **overlapping** answer categories in closed-ended questions.
- **Balance** rating scales.
- Have a **logical** structure.
- **Pilot!**

Questions?





Involving people:

Participant recruitment

What **activities** might involve participants?

- User research
 - Understand users (current and/or potential).
 - Gain insights into their practices, constraints, pain points, etc;
 - Understand the context, domain, etc...
- Ideation & design
 - e.g. in Participatory/Co-design
- Evaluation
 - Review conceptual design ideas/alternatives
 - Formative: Identify specific issues to fix, e.g. usability problems in prototypes.
 - Summative: How well does something perform? Metrics, benchmarking.

Studies / methods that involve participants

- Interviews
- Focus groups
- Direct observations (contextual inquiry in the field; usability testing—remote or lab-based)
- Diary studies;
- Ideation / Design workshops
- Questionnaires (e.g. targeting a specific population)
- ...

Who should you recruit?

- Depends on the project, but typically, **representative users**.
- Existing product or new product for existing user base:
 - **Existing personas** can be used to guide recruitment
 - If they don't cover all potential users, create some **new ones**.
- If there are no personas, make informed decisions about who to recruit, based on your understanding of the target domain and the brief.
- For design research do not *only* focus on **existing users** of a product.
 - Also look for **potential users** or users of competitor products, to provide a wider range of insights.



Overview of participant **recruitment process**

- Identify one or more **target groups**, based on existing personas and/or your team's understanding of the target domain.
- Decide **how many** participants to involve, for each target group.
- Define **eligibility criteria** (inclusion / exclusion).
- Prepare a **screener** questionnaire.
- Share screener questionnaire via distribution channels
 - e.g. social media, recruiters, special interest groups.
- **Review** screener responses and decide who to invite.
- Contact respondents, **inviting** them to participate in your study.
 - **Participant information sheet** will enable them to make an informed decision
- Arrange session **appointment** (date, time, location, requirements).

Recruitment **screener**

- Short questionnaire that potential participants complete, for you to assess their suitability as participants in a particular study.
- Questions are designed to determine whether respondents match target groups the researchers would like to engage with for a project, also meet other criteria for participation (e.g. minimum age).
- At the end of the screener questionnaire, use routing/branching based on the respondent's answers, to:
 - Inform those suitable for participation that their responses will be reviewed by researchers, and they may be contacted within a given period
 - *Politely* decline those unsuitable for participation.
- In both cases, remember to **thank them** for their interest in your study!

Recruitment **screener**: potential content

- **Welcome screen**: Provide key information about the study. Should include: confidentiality statement, study purpose, length, activities, incentive payment (if applicable), time period the study will run.
- **Demographic questions**: Important if you are designing a product targeting a particular demographic, or want to exclude certain people, e.g. under 18. Also useful for ensuring good distribution of ages, genders etc, if appropriate.
- **Behavioural questions**: Designed to distinguish between different user types (e.g. Which phone do you own? How often do you book flights? Where you do shop for groceries?
- **Availability questions**: If you already have dates in mind, you can potentially speed up scheduling by having participants indicate specific availability.
- **Contact information**: Enables you to contact respondents to invite them to participate in your study. (e.g. name, telephone number / email).
- **‘Thank you’ message**: At the end of the questionnaire, thank respondents and explain any next steps.

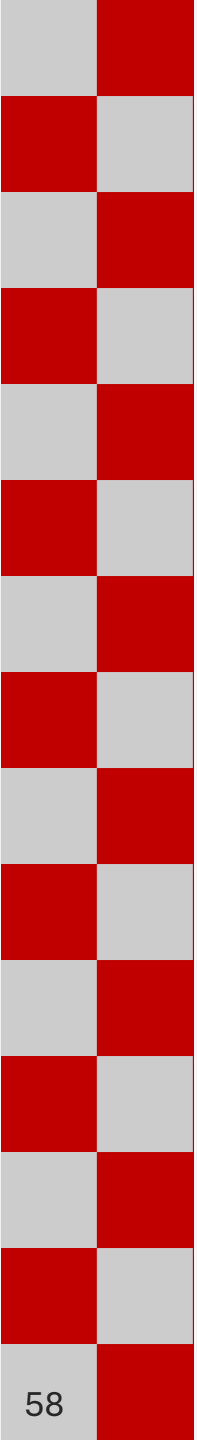
As always, make sure what you ask is **relevant** and **appropriate**, and keep the questionnaire as short as possible. Follow questionnaire design good practice.

Where to recruit participants

- Research providers (e.g. agencies like Saros, People for Research).
- Client's customer base.
- Social media.
- Within the organisation (internal products)
- Special interest groups.
- *Friends & family...?*

Questions?





Involving people

Ethics and Informed Consent

Informed consent and **ethical practice**

We have an ethical obligation towards the people we recruit as research participants

Human Subjects Protection:

When conducting research on human subjects, minimize harms and risks and maximize benefits; respect human dignity, privacy, and autonomy; take special precautions with vulnerable populations; and strive to distribute the benefits and burdens of research fairly.

What is Ethics in Research & Why is It Important? by David B. Resnik,
<http://www.niehs.nih.gov/research/resources/bioethics/whatis/>

Informed Consent and ethical practice

- We need to inform people about **what** they will be asked to do, what **data** we will collect and **how** we will use it, and **who** will have access to it.
- **Informed consent** should be obtained from all participants.
- Participants should not be subjected to **undue stress**
- The data should be **anonymised**: it should not be possible to identify individuals from written reports, images or video.
- City St George's University of London ethics information:
 - <https://www.citystgeorges.ac.uk/research/support/integrity-and-ethics/ethics>
 - <https://www.citystgeorges.ac.uk/research/support/integrity-and-ethics/guidance-and-resources>

Two main parts to Informed Consent

1. Informed

- Participants understand what is going to happen during data collection and with their data.
- Usually have a **Participant information sheet** giving details about what the study is about, what they can expect to happen, how their data is used and how it is kept safe.

2. Consent

- Participants have given *explicit* consent.
- Usually involves an **informed consent form** that is signed by both the participant and the researcher.

What participants should **know** in order to provide ***informed*** consent

- The purpose of the study.
- What participating in the study will entail.
- How the study will be recorded.
- How any other data will be collected
- What will happen to any data collected, including who will have access to it, and any confidentiality assured.
- Participants' rights, e.g. to withdraw from the study at any point.

For the coursework assignment, you will be provided with a **participant information sheet** and **informed consent form** that you can adapt.

Obtaining informed consent from participants

You must gain informed consent from a participant **before you capture any data**, including video/audio recording.

Before the session:

- Provide participants with a Participant Information sheet that explains the study.
Make sure they have enough time to read and understand it before agreeing to take part.

In the session:

- Check that the participant has read and understands the **Participant Information sheet**.
- Provide the participant with an **Informed Consent form**.
Ensure they understand what they are agreeing to.
- Ask if they have questions, e.g. about the study, session, etc—answer any they may have.
- Ask them to **sign** the form, to indicate they understand what they've been told, agree to you capturing and using specified data for the stated purposes, and are happy to participate.
- Once informed consent is granted, you can start recording and capturing data.

GDPR (General Data Protection Regulation)

- If you collect **personal data**, these data are subject to GDPR.
 - Data that can be directly / indirectly related to an identifiable individual.
- **Don't collect personal data unless you really have to.**
- **Don't send any personally identifiable information to any external service.**
 - e.g. anything that can potentially be used to identify a specific person, either on its own or when combined with another piece of information, like an email address, home address, birthdate, postcode, IP address...
- In collecting any information from participants, you must inform them of how the information will be used, and for how long it will be kept.
- The information must be used only for the stated purpose.

Questions?



Session 3 summary

- Query techniques such as interviews and questionnaires are very useful techniques for capturing data from participants during a design process.
 - They are easy to use, but it's not always easy to design them right.
 - There is a lot of guidance available on designing/using them.
- When you don't know much about something already, interviews will provide lots of rich data; questionnaires have limited value in early stage design.
- Human participants that we involve in our research must be treated ethically.
 - This isn't just good practice, it's the law.
 - Before collecting any data from participants, they must provide the researcher with their informed consent.